LEIC-T 2025/2026
Aprendizagem - Machine Learning
Homework 3
Deadline 15/10/2024 23:00
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I) Polynomial Regression (2 pts)

Consider a training set with 5 observations (sample) with dimension $D = 1$

$$x_1=1,\ x_2=0,\ x_3=-2,\ x_4=5,\ x_5=-5$$

With targets

$$t_1=-1,\ t_2=1,\ t_3=-1,\ t_4,=0,\ t_5=-1$$

Consider the basis function as well

$$\phi_j(x) = x^j$$

which can lead to a polynomial regression of the third degree

$$y(x, \mathbf{w}) = \sum_{j=0}^{3} w_j \cdot \phi_j(x) = w_0 + w_1 \cdot x + w_2 \cdot x^2 + w_3 \cdot x^3.$$

You can (should?) use your computer with Python, NumPy, MATLAB, Octave, Mathematica, etc. (whatever tool/language you like). Please indicate your results step by step.

(a) (0.5 pts)

Compute the design matrix Φ.

$$\Phi = \begin{pmatrix} \phi_0(\mathbf{x}_1) & \phi_1(\mathbf{x}_1) & \phi_2(\mathbf{x}_1) & \phi_3(\mathbf{x}_1) \\ \phi_0(\mathbf{x}_2) & \phi_1(\mathbf{x}_2) & \phi_2(\mathbf{x}_2) & \phi_3(\mathbf{x}_2) \\ \phi_0(\mathbf{x}_3) & \phi_1(\mathbf{x}_3) & \phi_2(\mathbf{x}_3) & \phi_3(\mathbf{x}_3) \\ \phi_0(\mathbf{x}_4) & \phi_1(\mathbf{x}_4) & \phi_2(\mathbf{x}_4) & \phi_3(\mathbf{x}_4) \\ \phi_0(\mathbf{x}_5) & \phi_1(\mathbf{x}_5) & \phi_2(\mathbf{x}_5) & \phi_3(\mathbf{x}_5) \end{pmatrix}$$

$$\Phi = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & -2 & 4 & -8 \\ 1 & 5 & 25 & 125 \\ 1 & -5 & 25 & -125 \end{pmatrix}$$

(b) (1.5 pts)

Compute the polynomial regression weights.

Solution:

By specifying as loss, the sum of squared errors, we can minimize it by computing the gradient and check where it is zero. Doing so, yields the closed form solution for the weights:

$$\mathbf{w} = \left(\mathbf{\Phi}^T \mathbf{\Phi}\right)^{-1} \mathbf{\Phi}^T \mathbf{T}$$

Applying it to the data we get:

$$\Phi^T \Phi = \begin{pmatrix} 5 & -1 & 55 & -7 \\ -1 & 55 & -7 & 1267 \\ 55 & -7 & 1267 & -31 \\ -7 & 1267 & -31 & 31\,315 \end{pmatrix}$$

$$(\Phi^T \Phi)^{-1} = \begin{pmatrix} 0.391759 & 0.0493362 & -0.0167806 & -0.00192518 \\ 0.0493362 & 0.275659 & -0.000891328 & -0.011143 \\ -0.0167806 & -0.000891328 & 0.00151361 & 0.0000338103 \\ -0.00192518 & -0.011143 & 0.0000338103 & 0.00048238 \end{pmatrix}$$

$$(\mathbf{\Phi}^T \mathbf{\Phi})^{-1} \mathbf{\Phi}^T = \begin{pmatrix} 0.42239 & 0.391759 & 0.241365 & -0.0217229 & -0.0337912 \\ 0.312961 & 0.0493362 & -0.416403 & 0.0124763 & 0.0416298 \\ -0.0161246 & -0.0167806 & -0.00921403 & 0.0208293 & 0.02129 \\ -0.012552 & -0.00192518 & 0.016637 & 0.00350267 & -0.00566251 \end{pmatrix}$$

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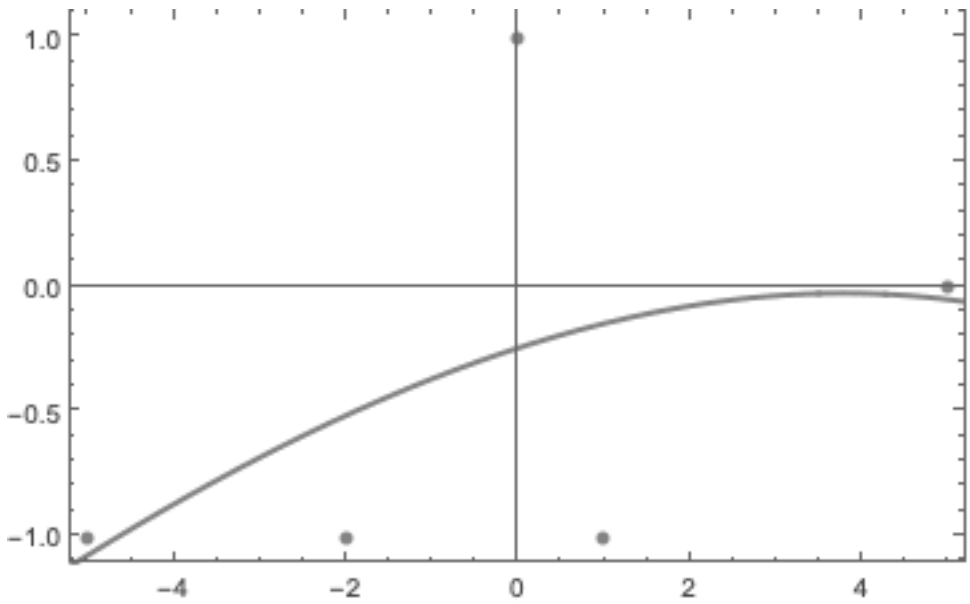
$T = (-1,1,-1,0,-1)^{\mathrm{T}}$

w = $(\Phi^{\mathrm{T}}\,\Phi)^{-1}\,\Phi^{\mathrm{T}}$ T= $(-0.2382047, 0.11115, -0.012732, -0.00034768)^{\mathrm{T}}$

$y = (-0.2382047 + \ 0.11115 * x - 0.012732 * x^2 - 0.00034768 * x^3$



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II) Perceptron (2pts)

Consider the following linearly separable training set:

$$x_1 = \begin{pmatrix} -2 \\ -2 \end{pmatrix}, x_2 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, x_3 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

with the target values

$$t_1=0,\ t_2=0,\ t_3=1$$

Initialize all weights to one (including the bias). Use a learning rate of one for simplicity.

Apply the perceptron learning algorithm with the sgn_0 activation function

$$sgn_0(x) = \begin{cases} 1 \text{ if } & x \geq 0 \\ 0 \text{ if } & x < 0 \end{cases}.$$

for **one** iteration.

Solution:

$$o_1=sgn_0=sgn_0\left(\left\langle \begin{matrix} 1 \\ 1 \\ 1 \end{matrix} \middle| \begin{matrix} 1 \\ -2 \\ -2 \end{matrix} \right\rangle\right) = 0, \qquad \delta_1 = (t_3 - o_3) = (0 - 0),$$

Wights do not change

$$o_2=sgn_0=sgn_0\left(\left\langle \begin{matrix} 1 \\ 1 \\ 1 \end{matrix} \middle| \begin{matrix} 1 \\ 1 \\ 0 \end{matrix} \right\rangle\right) = 1, \qquad \delta_1 = (t_3 - o_3) = (0 - 1) = -1,$$

$$\mathbf{w}=\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} - 1 * 1 * \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$o_3=sgn_0=sgn_0\left(\left\langle \begin{matrix} 0 \\ 0 \\ 1 \end{matrix} \middle| \begin{matrix} 1 \\ 0 \\ 1 \end{matrix} \right\rangle\right) = 1, \qquad \delta_1 = (t_3 - o_3) = (1 - 1) = 0,$$

Wights do not change

$$\mathbf{w}= \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

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III) Neural Network NN (4 pts)

Given the weights

$$W^{[1]} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ -1 & 1 & -1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 \end{pmatrix}$$
$$b^{[1]} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$
$$W^{[2]} = \begin{pmatrix} 0 & 0 & 0 \\ 2 & 0 & 0 \end{pmatrix}$$
$$b^{[2]} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

and the activation function ReLU of the hidden layer and SoftMax of the output layer using the cross-entropy error loss do a stochastic gradient descent update (with learning rate η = 1) for the training example:

$$\mathbf{x}=(1,1,1,1,1)^T \text{ and the target } \mathbf{t}=(1,0)^T,$$

Indicate the new weights and bias.

Solution:

$$\mathbf{z}^{[1]} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ -1 & 1 & -1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 5 \end{pmatrix}, \quad \mathbf{x}^{[1]} = \begin{pmatrix} 1 \\ 0 \\ 5 \end{pmatrix}$$

$$\mathbf{z}^{[2]} = \begin{pmatrix} 0 & 0 & 0 \\ 2 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 5 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \end{pmatrix},$$

$$\mathbf{x}^{[2]} = softmax \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} \exp(0) / (\exp(0) + \exp(2)) \\ \exp(2) / (\exp(0) + \exp(2)) \end{pmatrix} = \begin{pmatrix} 0.1192 \\ 0.8808 \end{pmatrix}$$

$$\delta^{[2]} = \frac{\partial E}{\partial \mathbf{x}^{[2]}} \circ \frac{\partial \mathbf{x}^{[2]}}{\partial \mathbf{z}^{[2]}} = \left(\mathbf{x}^{[2]} - \boldsymbol{t}\right) = \begin{pmatrix} 0.1192 \\ 0.8808 \end{pmatrix} - \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -0.880797 \\ 0.880797 \end{pmatrix}$$

$$\delta^{[1]} = \left(\frac{\partial \mathbf{z}^{[2]}}{\partial \mathbf{x}^{[1]}}\right)^T \cdot \delta^{[2]} \circ \frac{\partial \mathbf{x}^{[1]}}{\partial \mathbf{z}^{[1]}} = \mathbf{W}^{[2]^T} \cdot \delta^{[2]} \circ sgn_0\left(\mathbf{z}^{[1]}\right) = \mathbf{W}^{[2]^T} \cdot \delta^{[2]} \circ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} =$$

$$= \left(\begin{pmatrix} 0 & 2 \\ 0 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} -0.880797 \\ 0.880797 \end{pmatrix} \right) \circ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1.761594 \\ 0 \\ 0 \end{pmatrix}$$

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$$\frac{\partial E}{\partial \mathbf{W}^{[1]}} = \delta^{[1]} \frac{\partial \mathbf{z}^{[1]}}{\partial \mathbf{W}^{[1]}} = \delta^{[1]} (\mathbf{x}^{[0]})^T = \begin{pmatrix} 1.761594 \\ 0. \\ 0. \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1.761594 & 1.761594 & 1.761594 & 1.761594 & 1.761594 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Change in weights of $W^{[1]}$ and in bias.

$$\mathbf{W}^{[1]} = \mathbf{W}^{[1]} - \eta \frac{\partial E}{\partial \mathbf{W}^{[1]}} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ -1 & 1 & -1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 \end{pmatrix} - 1 * \begin{pmatrix} 1.761594 & 1.761594 & 1.761594 & 1.761594 & 1.761594 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} =$$

$$= \begin{pmatrix} -0.761594 & -1.761594 & -1.761594 & -1.761594 & -1.761594 \\ -1 & 1 & -1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 \end{pmatrix}$$

$$\frac{\partial E}{\partial \mathbf{b}^{[1]}} = \delta^{[1]} \frac{\partial \mathbf{z}^{[1]}}{\partial \mathbf{b}^{[1]}} = \delta^{[1]} = \begin{pmatrix} 1.761594 \\ 0 \\ 0 \end{pmatrix}$$

$$\mathbf{b}^{[1]} = \mathbf{b}^{[1]} - \eta \frac{\partial E}{\partial \mathbf{b}^{[1]}} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} - 1 * \begin{pmatrix} 1.761594 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} -1.761594 \\ 0 \\ 0 \end{pmatrix}$$

$$\frac{\partial E}{\partial \mathbf{W}^{[2]}} = \delta^{[2]} \frac{\partial \mathbf{z}^{[2]}}{\partial \mathbf{W}^{[2]}} = \delta^{[2]} (\mathbf{x}^{[1]})^T = \begin{pmatrix} -0.880797 \\ 0.880797 \end{pmatrix} (1,0,5) = \begin{pmatrix} -0.880797 & 0 & -4.403985 \\ 0.880797 & 0 & 4.403985 \end{pmatrix}$$

Change in weights of $W^{[2]}$ and in bias.

$$\mathbf{W}^{[2]} = \mathbf{W}^{[2]} - \eta \frac{\partial E}{\partial \mathbf{W}^{[2]}} = \begin{pmatrix} 0 & 0 & 0 \\ 2 & 0 & 0 \end{pmatrix} - 1 * \begin{pmatrix} -0.880797 & 0 & -4.403985 \\ 0.880797 & 0 & 4.403985 \end{pmatrix} = \begin{pmatrix} 0.880797 & 0 & 4.403985 \\ 1.119203 & 0 & -4.403985 \end{pmatrix}$$

$$\frac{\partial E}{\partial \mathbf{b}^{[2]}} = \delta^{[2]} \frac{\partial \mathbf{z}^{[2]}}{\partial \mathbf{b}^{[2]}} = \delta^{[2]} = \begin{pmatrix} -0.880797 \\ 0.880797 \end{pmatrix}$$

$$\mathbf{b}^{[2]} = \mathbf{b}^{[2]} - \eta \frac{\partial E}{\partial \mathbf{b}^{[2]}} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} - 1 * \begin{pmatrix} -0.880797 \\ 0.880797 \end{pmatrix} = \begin{pmatrix} 0.880797 \\ -0.880797 \end{pmatrix}$$